Double-Angle Formulas and Trigonometric Equations

ANSWER KEY

Double-angle formulas

1 Given that $\cos\theta = -\frac{3}{5}$ and θ lies in Quadrant II, find:

a $\sin 2\theta \;= -\frac{24}{25}$ **b** $\cos 2\theta \;= -\frac{7}{25}$ **c** $\tan 2\theta \;= \frac{24}{7}$

2 Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

$\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$.

Solving trigonometric equations

3 Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

$\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.

4 Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

Factor: $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

5 Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.

$2\sin x\cos x - \sin x = 0$, so $\sin x(2\cos x - 1) = 0$. From $\sin x = 0$: $x = 0, \pi$; from $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}, \frac{5\pi}{3}$. Solution set: $\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.

6 Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.

$\tan x = \pm\sqrt{3}$, so $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

Half-angle formulas

7 Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

$\cos 15° = \sqrt{\frac{1+\cos 30°}{2}} = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$ (positive since $15°$ is in Quadrant I).

8 Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

$\sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = \sqrt{\frac{18/25}{2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$ (positive since $\frac{\theta}{2}$ is in Quadrant I).

Equations requiring double-angle substitution

9 Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x + \sin x = 0$, so $2\sin^2 x - \sin x - 1 = 0$: $(2\sin x + 1)(\sin x - 1) = 0$. From $\sin x = -\frac{1}{2}$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$; from $\sin x = 1$: $x = \frac{\pi}{2}$. Solution set: $\left\{\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\right\}$.

10 Solve $\sin 2x - \cos x = 0$ on the interval $[0, 2\pi)$.

$2\sin x\cos x - \cos x = 0$, so $\cos x(2\sin x - 1) = 0$. From $\cos x = 0$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$; from $\sin x = \frac{1}{2}$: $x = \frac{\pi}{6}, \frac{5\pi}{6}$. Solution set: $\left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}$.

Applications of double-angle formulas

11 A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.

$R(\theta) = \frac{v^2}{g}\sin 2\theta$ is maximized when $\sin 2\theta = 1$, i.e. $2\theta = 90°$, so $\theta = 45°$.

12 Given $\sin\theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

$\cos\theta = \sqrt{1 - 1/9} = \frac{2\sqrt{2}}{3}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2}{9} = \frac{7}{9}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{9}$.

More equation practice

13 Solve $4\sin^2 x - 3 = 0$ on the interval $[0, 2\pi)$.

$\sin^2 x = \frac{3}{4}$, so $\sin x = \pm\frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{3}$, giving $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

14 Solve $\cos 2x = \cos x$ on the interval $[0, 2\pi)$, using $\cos 2x = 2\cos^2 x - 1$.

$2\cos^2 x - 1 = \cos x$: $2\cos^2 x - \cos x - 1 = 0$, so $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

Half-angle formulas for tangent

15 State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

$\tan\frac{\theta}{2} = \frac{\sin\theta}{1 + \cos\theta}$. With $\theta = 30°$: $\tan 15° = \frac{\sin 30°}{1 + \cos 30°} = \frac{1/2}{1 + \sqrt{3}/2} = \frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$ (after rationalizing).

Equations with multiple angle types

16 Solve $\sin 2x + \sin x = 0$ on $[0, 2\pi)$.

$2\sin x\cos x + \sin x = 0 \Rightarrow \sin x(2\cos x + 1) = 0$. From $\sin x = 0$: $x = 0, \pi$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$. Solution set: $\left\{0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}\right\}$.

17 Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x = 1 - 3\sin x \Rightarrow -2\sin^2 x + 3\sin x = 0 \Rightarrow \sin x(3 - 2\sin x) = 0$. From $\sin x = 0$: $x = 0, \pi$. From $\sin x = \frac{3}{2}$: no solution (out of range). Solution set: $\{0, \pi\}$.

Triple-angle preview

18 Using $\sin(2x+x)=\sin 2x\cos x+\cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x=3\sin x-4\sin^3 x$.

$\sin 3x=\sin 2x\cos x+\cos 2x\sin x=(2\sin x\cos x)\cos x+(1-2\sin^2 x)\sin x=2\sin x\cos^2 x+\sin x-2\sin^3 x.$
Substitute $\cos^2 x=1-\sin^2 x$:
$=2\sin x(1-\sin^2 x)+\sin x-2\sin^3 x=2\sin x-2\sin^3 x+\sin x-2\sin^3 x=3\sin x-4\sin^3 x.$

Graphical verification of a double-angle identity

19 The functions $y=\cos 2x$ and $y=1-2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.

The two curves overlapping perfectly (indistinguishable) is a visual confirmation of the identity $\cos 2x=1-2\sin^2 x$ — since both functions produce the exact same output for every input x, their graphs coincide everywhere.

Solving equations restricted to a smaller interval

20 Solve $2\cos 2x-1=0$ for $x\in[0,\pi)$ only (note the restricted domain).

$\cos 2x=\frac{1}{2}$: over $[0,2\pi)$ for $2x$, solutions are $2x=\frac{\pi}{3},\frac{5\pi}{3},\frac{7\pi}{3},\frac{11\pi}{3}$, giving $x=\frac{\pi}{6},\frac{5\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6}$. Restricting to $x\in[0,\pi)$: only $x=\frac{\pi}{6},\frac{5\pi}{6}$ qualify.